

Picard-Fuchs equations and mirror maps for hypersurfaces

Source-backed arXiv sample

1 Extracted Lists

The list environments below are extracted from arXiv LaTeX source and wrapped for pdfLaTeX validation.

1. If we analytically continue along a simple loop around $z = 0$ in the counterclockwise direction, t becomes $t + 1$. (It will be convenient to also introduce $q = e^{2\pi it}$, which remains single-valued near $z = 0$.)
2. There are cycles γ_0 and γ_1 such that $\int_{\gamma_0} \omega(z)$ is single valued near $z = 0$, and

$$t = \frac{\int_{\gamma_1} \omega(z)}{\int_{\gamma_0} \omega(z)}$$

in an angular sector near $z = 0$.

1. There is a period function for $\omega(z)$,

$$f_0(z) := \int_{\gamma_0} \omega(z)$$

which is single-valued near $z = 0$. This period function is unique up to multiplication by a constant. (This implies that the cycle γ_0 is also unique up to a constant multiple.)

In particular, the family of meromorphic n -forms

$$\tilde{\omega}(z) := \frac{\omega(z)}{\int_{\gamma_0} \omega(z)}$$

will have the property that

$$\int_{\gamma} \tilde{\omega}(z) \equiv 1$$

for some γ , and it is the unique such family up to constant multiple.

2. Fixing a choice of period function $f_0(z)$ as in part (1), there is a period function

$$f_1(z) := \int_{\gamma_1} \omega(z)$$

such that $\varphi(z) := f_1(z)/f_0(z)$ transforms as

$$\varphi(z) \mapsto \varphi(z) + 1$$

upon transport around $z = 0$ in the counterclockwise direction. The ratio $\varphi(z)$ is unique up to the addition of a constant.